

EDUCATION

Master of Science, Information Technology, Arizona State University, 4.0 August 2022 — May 2024

SKILLS

- Languages:** Python, SQL, C, Java, JavaScript/TypeScript, HTML, CSS
- Web Development:** React, Next.js, Angular, Flask, Django, FastAPI, Node.js, NestJS, Express.js, React Native, Expo, RESTful APIs, GraphQL, WebSockets, gRPC, Apollo Client
- Cloud Services:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, Vercel
- Databases:** DynamoDB, MySQL, PostgreSQL, MongoDB, Elasticsearch, SQLite, Apache Spark, BigQuery, Hasura
- Data Science & ML:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Apache Spark, PySpark, LangChain, MATLAB, Gen AI
- Data Analytics:** Microsoft Excel, Tableau, Looker, Power BI, Google Analytics
- Tools & Testing:** PyTest, Gherkin, Git, Visual Studio, Docker, BeautifulSoup, Kafka, ETL, Agile, JIRA, Kubernetes, Redis, Jest, Playwright, Terraform, Prometheus, Grafana, GitHub Actions, Microservices
- Management:** Agile Methodologies, JIRA, Project Management, ETL, SDLC

EXPERIENCE

Software Developer(Part-time), Arasan Chip Systems Inc, San Jose, USA June 2025 – Present

- Developed and integrated a Slack bot into client’s Next.js application using Vercel’s AI SDK, streamlining client AI workflows and driving a 10% uplift in revenue.
- Optimized client’s AI-driven query workflow by integrating a rate limiter and adding social media connectors (Gmail, Discord, etc.), enabling users to query important emails and conversations, boosting user engagement by 20%.

AI/ML Backend Developer, Arizona State University, Remote, USA Aug 2024 – Jun 2025

- Spearheaded an AI-powered profile verification microservice for 10K+ users using Face Recognition API, distance-based scoring, React Native, Flask, and GCP Functions boosting fraud detection accuracy by 25%.
- Engineered a scalable image processing pipeline with Multer, GCP Storage, and Hasura GraphQL, enabling secure storage and retrieval of profile images, reducing verification latency by 30%.
- Containerized a FastAPI service for OTLP JSON span ingestion and implemented Redis Streams XGROUP consumer groups to ensure at-least-once delivery.
- Developed asynchronous RCA processors with Python asyncio calling the Google Gemini-2 API to generate concise root-cause insights, and upserting results into Supabase for seamless end-to-end fault-diagnosis workflows.
- Led a 3 engineer team and coordinated cross-functional sprints to deliver a full-stack AI healthcare platform using React/Vite & Tailwind CSS frontend, Node.js, Drizzle ORM on NeonDB backend, Google GenAI API integration, and Docker CI/CD.

Software Developer(Contract), India July 2020 – July 2022

- Developed onboarding flows using React and Material UI, improving user retention by 30% through better UX and seamless registration.
- Reduced server costs by 15% by optimization through AWS ELB with weighted round-robin distribution, handling traffic spikes efficiently.
- Optimized 10 TB/day PySpark ETL on AWS EMR & S3, reducing batch-to-dashboard latency by 70% and enabling near-real-time analytics.
- Architected and scaled a fault-tolerant Django, PostgreSQL API in Docker on AWS Fargate to handle 10 K concurrent sessions at ≤30 ms latency with 99.9% uptime.
- Built an ELK log-analytics pipeline ingesting 50 K+ logs per release via Logstash into Elasticsearch & Kibana dashboards, cutting defect-triage time by 60%.

Software Developer Intern, Indian Institute of Technology Guwahati, India May 2019 - May 2020

- Developed a HIPAA compliant Pydantic validation library for ETL jobs that auto-detects schema drift and masks PHI, reducing data defects by 70% in a 5 TB healthcare analytics pipeline.
- Led a team of three to design and train an LSTM-based Seq2Seq translation model on 50K+ bilingual text pairs, applying Optuna for hyperparameter tuning to achieve 88% accuracy.
- Built AWS CI/CD pipelines with SageMaker, CodePipeline, Lambda, S3, and EC2 to automate ML model training, versioning, and deployment halving iteration cycles.

PROJECTS

Social Media Platform

Designed and developed a scalable full-stack social media platform using Next.js App Router, TypeScript, and Prisma with PostgreSQL. Integrated Clerk for secure authentication and role-based authorization. Implemented server-side data fetching via Dynamic/Static Routes and API route handlers for efficient revalidation and caching. Styled with Tailwind CSS and Shadcn UI, and added file upload functionality using UploadThing. Enhanced real-time performance by integrating Kafka for event-driven architecture and Redis for low-latency content delivery and caching.

Multi-Agent RAG Chatbot

Architected a production-grade multi-agent Retrieval-Augmented Generation (RAG) chatbot using CrewAI, combining Qdrant vector search, semantic chunking, and multi-agent task planning for accurate and contextual responses. Built a dynamic document processing pipeline with Chonkie, indexing over 200+ technical docs, reducing retrieval latency by 40% and minimizing manual input by 70%. Integrated evaluation metrics with Gemini, improving overall response relevancy by 25% and aligning outputs with industry-standard accuracy requirements.